Operator metering: This enables usage reporting for operators that provide specialised services.

Introducing Kubevirt

Kubevirt has been developed as a virtual machine management add-on for Kubernetes, to address the need of development teams to adopt Kubernetes with existing virtual machine-based workloads that cannot be easily containerised. It provides a unified development platform whereby developers can build, modify and deploy applications residing in both application containers as well as virtual machines, in a common, shared environment.

Kubevirt extends Kubernetes by adding virtualisation resource types (especially of the VM type) through the Kubernetes Custom Resource Definitions API. It provides the ability to manage additional resources alongside the resources provided by Kubernetes, by default. But the resources themselves are not enough to launch virtual machines; so, instead, the functionality and business logic is added to the cluster by running additional controllers and agents on an existing cluster.

The architecture of Kubevirt

Kubevirt is built using a service-oriented architecture and a choreography pattern. It provides additional functionality to your Kubernetes cluster to perform virtual machine management. You may be aware of how pods are created in Kubernetes from specifications, and how controllers perform the necessary actions to make the pod alive to match the required state. Kubevirt uses similar mechanisms with additional types added to the Kubernetes API, additional controllers for business logic and additional daemons for node-specific logic. Figure 1 illustrates the initial architecture of Kubevirt with its additional controllers, daemons and their communication.

Kubevirt is deployed on top of a Kubernetes cluster, which allows users to run Kubernetes-native workloads next to the VMs managed through Kubevirt.

Components of Kubevirt

Kubevirt consists of a set of services as described in the architecture given below.

virt-api-server serves as the entry point for all virtualisation related flows and takes care to update the virtualisation related custom resource definition (CRD). It also acts as the main entry point to Kubevirt, and is responsible for the defaulting and validation of the provided VMs.

virt-controller provides all the cluster-wide virtualisation functionality. It is responsible for monitoring the VM (CRDs) and managing the associated pods. The virt-controller is responsible for creating and managing the life cycle of the pods associated to the VM objects.

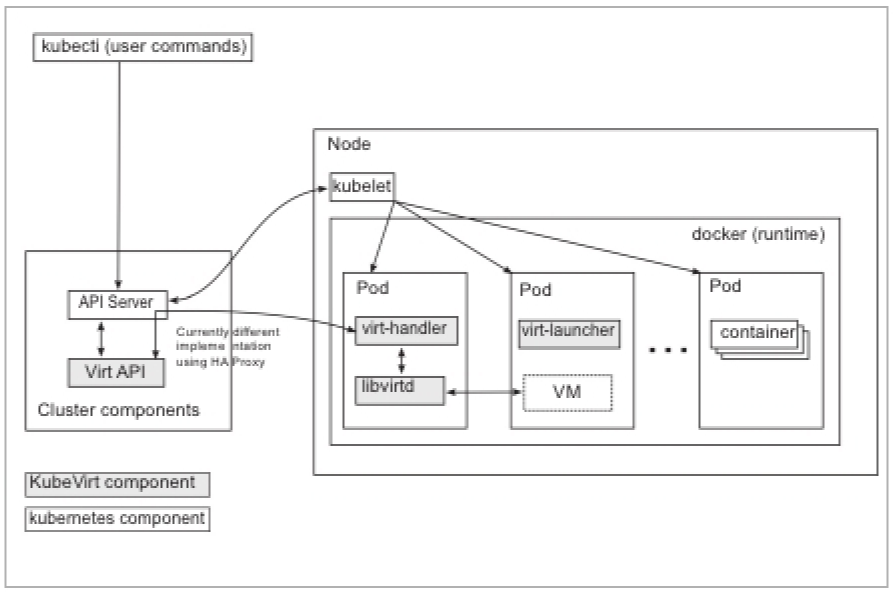

Figure 1: Kubevirt architecture

virt-handler is a Kubernetes DaemonSet needed on every host in the cluster. It performs functions similar to those of Kubelet, where it is reactive and watching out for changes in the VM object. Once changes are detected, it will perform all the necessary operations to change a VM to meet the required state.

virt-launcher is responsible for providing the cgroups and namespaces, which will be used to host the VM process. virt-handler signals virt-launcher to start a VM by passing the VM's CRD object to virt-launcher.

libvirtd instance is present in every VM Pod. virt-launcher uses libvirtd to manage the life cycle of the VM process.

Similar to Kubernetes, Kubevirt also has support for providing additional functionality for storage and networking, with additional controllers and/or plugins.

Kubevirt community

Kubevirt is a work in progress. The code base is hosted on GitHub and is distributed under the Apache License, version 2.0. The contributors hang out on IRC at #Kubevirt at FreeNode. You can join the Kubevirt-dev Google Group for developer discussions. END

References

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[2] *https://kiwiirc.com/client/irc.freenode.net/kubevirt*
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